

Topic C9

# Metals

**Concept guide · Worked examples · Cambridge-style practice · Fully explained answers**

## What this guide teaches

A complete explanation of the syllabus concepts, chemical reasoning, worked applications, common misconceptions and original exam-style practice for this topic.

## Syllabus map

|      |                          |
|------|--------------------------|
| C9.1 | Properties and uses      |
| C9.2 | Alloys                   |
| C9.3 | Reactivity series        |
| C9.4 | Extraction of metals     |
| C9.5 | Corrosion and prevention |

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

# 1. Properties and uses

Metals conduct heat and electricity, are malleable and ductile, and often have high density and melting point. Uses follow properties: copper wiring combines electrical conductivity and ductility; aluminium aircraft parts combine low density, strength and corrosion resistance. Chemical properties vary widely with reactivity.

## Examiner insight

Link each named property explicitly to the requirement of the use.

# 2. Alloys

An alloy is a mixture containing a metal and other elements. Different-sized atoms disrupt regular layers in the metal lattice, so layers slide less easily and the alloy is usually harder and stronger than the pure metal. Stainless steel contains iron with chromium and often nickel; brass contains copper and zinc.

## Examiner insight

Alloys are mixtures, not compounds; their compositions can vary.

# 3. Reactivity series

A useful order is K, Na, Ca, Mg, Al, C, Zn, Fe, H, Cu, Ag, Au. A more reactive metal displaces a less reactive metal from its compound. Metals above hydrogen react with dilute acids to release hydrogen. Reactions with water or steam also reveal reactivity.

## Worked example

$\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$  because zinc is more reactive than copper.

## Examiner insight

Use observations such as coating, colour change and temperature change as evidence of displacement.

# 4. Extraction

Very unreactive metals may occur native. Metals below carbon are commonly extracted by carbon reduction; metals above carbon require electrolysis of molten compounds. Extraction method balances reactivity, energy cost and ore availability. Recycling usually saves energy and reduces mining and waste.

## Worked example

$\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ ; iron oxide is reduced by carbon monoxide.

## Examiner insight

Aluminium extraction uses electrolysis despite high cost because aluminium oxide cannot be reduced by carbon.

## 5. Rusting and corrosion

Rusting of iron requires both oxygen and water and produces hydrated iron(III) oxide. Barrier methods (paint, oil, plastic) exclude oxygen/water. Galvanising coats iron with zinc, which also gives sacrificial protection because zinc is more reactive. Stainless steel resists corrosion through alloying.

### Examiner insight

A scratched zinc coating still protects iron sacrificially; a scratched tin coating may accelerate iron corrosion.

## Exam technique

### High-value habits

Use precise chemical vocabulary, balance equations, show every calculation step, include units and state symbols where required, distinguish observations from conclusions, and answer the command word directly.

## Retrieval checklist

- Recall all definitions, tests and equations without notes.
- Explain one unfamiliar observation using particles, bonding or collision theory.
- Check charge, atoms, units and significant figures in every calculation.
- Complete one timed set and mark it only after finishing.

# Practice Paper

Topic C9

**etals****Instructions**

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

**Total marks: 50**

Suggested time: 60 minutes.

## Section A · Multiple choice

Choose the best answer for each question.

1. Why is copper used for electrical wiring? [1]
  - A low conductivity
  - B conductivity and ductility
  - C brittleness
  - D high reactivity
  
2. Why are alloys often harder than pure metals? [1]
  - A They contain molecules.
  - B Different-sized atoms hinder layer sliding.
  - C They have no electrons.
  - D All atoms become identical.
  
3. Which metal reacts most vigorously with cold water? [1]
  - A copper
  - B iron
  - C sodium
  - D silver
  
4. Which metal displaces copper from copper sulfate? [1]
  - A silver
  - B gold
  - C zinc
  - D platinum
  
5. Which metal does not release H<sub>2</sub> from dilute acid? [1]
  - A magnesium
  - B zinc
  - C iron
  - D copper
  
6. Which metal is extracted by electrolysis? [1]
  - A iron
  - B copper
  - C aluminium
  - D silver
  
7. In  $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ , Fe<sub>2</sub>O<sub>3</sub> is [1]
  - A oxidised
  - B reduced
  - C a catalyst
  - D unchanged

8. Rusting requires [1]
- A oxygen only
  - B water only
  - C oxygen and water
  - D carbon dioxide only
9. Sacrificial protection uses a metal that is [1]
- A less reactive than iron
  - B more reactive than iron
  - C a noble gas
  - D non-conducting
10. Which is an environmental benefit of recycling aluminium? [1]
- A More mining
  - B Higher energy use
  - C Less ore extraction and energy use
  - D More landfill

## Section B · Structured questions

11 Copper is used in electrical cables and aluminium in aircraft.

(a) Give two properties making copper suitable. [2]

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(b) Give two properties making aluminium suitable. [2]

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(c) Explain metallic conductivity. [2]

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(d) Explain metallic malleability. [2]

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**[Total: 8]**

12 A student adds zinc to copper(II) sulfate solution.

(a) State two observations. [2]

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(b) Write the symbol equation. [2]

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(c) Explain the reaction using reactivity. [2]

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(d) Identify what is oxidised and reduced. [2]

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**[Total: 8]**

13 Iron is extracted in a blast furnace using carbon monoxide.

(a) Write the reduction equation for iron(III) oxide. [2]

(b) Explain why this is reduction. [2]

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(c) Explain why aluminium is not extracted this way. [2]

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(d) Give two benefits of metal recycling. [2]

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**[Total: 8]**

14 Iron nails are placed in different conditions to investigate rusting.

(a) State the two substances required. [2]

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(b) Describe a control showing oxygen is required. [2]

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(c) Explain how paint prevents rusting. [2]

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(d) Explain sacrificial protection by zinc. [2]

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**[Total: 8]**

15 Pure aluminium is compared with an aluminium alloy.

(a) Define alloy. [2]

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(b) Describe the structure of pure aluminium. [2]

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(c) Explain why the alloy is harder. [3]

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(d) State one disadvantage of alloying. [1]

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**[Total: 8]**

# Worked solutions

**Remove or fold back this section while completing the practice paper.**

## Marking routine

Mark each part separately. Award method and chemistry marks where shown, then identify the first point at which an answer becomes incorrect.

## Solutions · Multiple choice

- 1 B. It carries charge and can be drawn into wires.
- 2 B. Lattice distortion blocks easy movement of layers.
- 3 C. Sodium is high in the reactivity series.
- 4 C. Zinc is more reactive than copper.
- 5 D. Copper is below hydrogen.
- 6 C. Aluminium is above carbon.
- 7 B. It loses oxygen.
- 8 C. Both are necessary.
- 9 B. It oxidises in preference to iron.
- 10 C. Recycling avoids energy-intensive extraction and conserves ore.

## Solutions · Structured questions

11. 11(a) Electrical conductivity and ductility [2]. (b) Low density and strength/corrosion resistance [2]. (c) Delocalised electrons move and carry charge [2]. (d) Ion layers slide while metallic attraction remains [2].
12. 12(a) Blue fades; reddish-brown copper deposits; zinc dissolves/temperature rises [any 2]. (b)  $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$  [2]. (c) Zinc is more reactive and displaces copper [2]. (d) Zn oxidised,  $\text{Cu}^{2+}$  reduced [2].
13. 13(a)  $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$  [2]. (b) Iron oxide loses oxygen [2]. (c) Aluminium is more reactive than carbon, so oxide cannot be reduced by carbon [2]. (d) Less energy, mining, waste or habitat damage [any 2].
14. 14(a) Oxygen and water [2]. (b) Boiled water under oil excludes oxygen and nail does not rust [2]. (c) Barrier prevents oxygen/water reaching iron [2]. (d) Zinc is more reactive and oxidises instead of iron [2].
15. 15(a) Mixture of a metal with other elements [2]. (b) Regular positive-ion lattice with delocalised electrons [2]. (c) Different-sized atoms distort layers so they cannot slide easily [3]. (d) May reduce conductivity/ductility or increase cost [1].

### Self-assessment

Circle the first incorrect step and classify it as knowledge, equation or formula, mole ratio, unit conversion, observation, practical method or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.